

BST Vol 7 Ch 8 — Lagrangian + Hamiltonian EM (v0.4.1, substantive 3-level pedagogy upgrade)

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Vol 7 Chapter 8 — Lagrangian + Hamiltonian EM

Headline result

The Lagrangian formulation of electromagnetism is the foundation of relativistic field theory + QED + Standard Model:

$$\mathcal{L}_{EM} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + J^\mu A_\mu$$

Euler-Lagrange equations on this Lagrangian yield Maxwell's equations. Hamiltonian formulation:

$$\mathcal{H}_{EM} = \frac{1}{2}(E^2 + B^2) - J \cdot A$$

provides energy density + canonical conjugate momenta $\pi^i = -E^i$ for canonical quantization $\rightarrow$ QED.

BST identification: this is the substrate Yang-Mills action on Bergman line bundle $L_\lambda \rightarrow D_{IV}^5$ (T2477 RIGOROUSLY VERIFIED Saturday SP-31 #286). Gauge invariance $A_\mu \rightarrow A_\mu + \partial_\mu \Lambda$ preserved.

Substrate mechanism

Yang-Mills action substrate-natural form (T2477):

Per Lyra Saturday SP-31 #286: gauge field A_μ is a connection 1-form on Bergman line bundle $L_\lambda \rightarrow D_{IV}^5$ with structure group $K = SO(5) \times SO(2)$. $U(1)_{em}$ is an abelian sub-bundle. Field strength $F = dA + A \wedge A$ reduces to $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ for $U(1)$.

Yang-Mills action:

$$S_{YM} = -\frac{1}{4} \int_{\mu\nu} F^{\mu\nu} d^4x$$

substrate-natural quadratic functional of curvature.

Coupling to matter (sources J):

For charged matter current J^μ (Vol 7 Ch 2-3), full Lagrangian adds source term:

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + J^\mu A_\mu$$

Gauge invariance:

Under $A_\mu \rightarrow A_\mu + \partial_\mu \Lambda$: $-F_{\mu\nu} \rightarrow F_{\mu\nu}$ (gauge invariant) - $J^\mu A_\mu \rightarrow J^\mu A_\mu + J^\mu \partial_\mu \Lambda = J^\mu A_\mu + \partial_\mu (J^\mu \Lambda) - (\partial_\mu J^\mu) \Lambda$

The boundary term vanishes; remaining term vanishes by current conservation $\partial_\mu J^\mu = 0$ (Vol 8 Ch 5 + T2475). Gauge invariance + current conservation are dual constraints.

Hamiltonian formulation:

Canonical conjugate momenta:

$$\pi^i = \frac{\partial \mathcal{L}}{\partial(\partial_0 A_i)} = -F^{0i} = -E^i$$

Hamiltonian density:

$$\mathcal{H} = \pi^i \partial_0 A_i - \mathcal{L} = \frac{1}{2} (E^2 + B^2) - J \cdot A$$

First term: EM field energy density. Second term: source-field coupling.

Gauge fixing:

A_μ has gauge redundancy $\rightarrow$ fix gauge for quantization: - Coulomb gauge: $\nabla \cdot A = 0$ (non-Lorentz-covariant; useful for non-relativistic) - Lorenz gauge: $\partial_\mu A^\mu = 0$ (Lorentz-covariant; standard for QED) - Axial gauge: $A_3 = 0$ (light-front; useful for QCD)

Match precision

D-tier on substrate Yang-Mills framework (T2477 RIGOROUSLY VERIFIED). Classical EM Lagrangian/Hamiltonian preserved at any precision. QED corrections systematically derivable from canonical quantization.

Cross-volume dependencies

- Vol 1 Ch 8 (Gauge Theory + T2477) — substrate Yang-Mills framework
- Vol 7 Ch 2 (Maxwell) — equations derived from this Lagrangian
- Vol 7 Ch 7 (Relativistic EM) — covariant formulation
- Vol 2 Ch 8 (Coupling Constants) — α renormalization via canonical quantization
- Vol 8 Ch 3 (Lagrangian Mechanics) — particle-mechanics version
- Vol 8 Ch 4 (Hamiltonian Mechanics) — particle-mechanics analog of canonical structure
- Vol 8 Ch 5 (Symmetries + Noether) — T2475 current conservation

K-audit anchor

K224 Vol 7 Ch 8 Lagrangian EM K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1 — Bright 5th-grader

Physicists like to derive everything from one master equation called the Lagrangian. Once you have the Lagrangian, all the laws of motion come out by applying a math trick called “variation” (asking which configurations make the Lagrangian’s total be at a minimum or maximum).

For electromagnetism, the Lagrangian is:

$$\mathcal{L} = -\frac{1}{4}F^2 + J \cdot A$$

(F is the field strength, A is the potential, J is the charge current). Apply variation $\rightarrow$ out come Maxwell’s 4 equations! Beautiful.

BST predicts: this Lagrangian comes from substrate D_IV⁵. The photon is a “connection” on substrate’s Bergman bundle, and the Yang-Mills action is the substrate-natural way to encode photon dynamics.

Level 2 — Undergraduate physics student

Lagrangian formulation of EM:

The EM Lagrangian density:

$$\mathcal{L}_{EM} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + J^\mu A_\mu$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the field strength tensor.

Euler-Lagrange equations on A_μ :

$$\frac{\partial \mathcal{L}}{\partial A_\mu} - \partial_\nu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\nu A_\mu)} \right) = 0$$

Compute: $\partial \mathcal{L} / \partial A_\mu = J^\mu$; $\partial \mathcal{L} / \partial (\partial_\nu A_\mu) = -F^{\nu\mu} = F^{\mu\nu}$. Therefore:

$$\partial_\nu F^{\nu\mu} = J^\mu$$

This is the source Maxwell equations (Gauss-E + Ampère-Maxwell). The Bianchi identity $\partial_{[\mu} F_{\nu\rho]} = 0$ (geometric identity) gives source-free Maxwell equations.

Hamiltonian formulation:

Canonical conjugate to A_i : $\pi^i = \partial \mathcal{L} / \partial (\partial_0 A_i) = -E^i$.

Hamiltonian density: $H = \frac{1}{2}(E^2 + B^2) - J \cdot A$.

Gauge invariance + conservation: - $A_\mu \rightarrow A_\mu + \partial_\mu \Lambda$ leaves $F_{\mu\nu}$ unchanged - Current J^μ must satisfy $\partial_\mu J^\mu = 0$ (conservation) for gauge invariance of action

Gauge fixing for quantization: - Lorenz gauge $\partial_\mu A^\mu = 0$: covariant, standard for QED - Coulomb gauge $\nabla \cdot A = 0$: useful for atomic physics - Axial gauge $A_3 = 0$: useful for QCD

BST framework: - T2477 (Lyra Saturday): gauge field A_μ as Bergman bundle connection - Yang-Mills action substrate-natural for any connection on bundle - Canonical quantization $\rightarrow$ QED (Vol 2 Ch 8)

Level 3 — Graduate physics student / theorem-level

Substrate Yang-Mills framework (T2477 RIGOROUSLY VERIFIED):

Bergman line bundle $L_\lambda \rightarrow D_{IV}^5$ has structure group $K = SO(5) \times SO(2)$. Connection $\omega \in \Omega^1(L_\lambda; \mathbb{R})$. Local representation $A_\mu = \omega(\partial_\mu)$. Curvature 2-form $F = d\omega + \omega \wedge \omega$. For abelian $U(1)$ _em sub-bundle: $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

Action principle:

$$S = -\frac{1}{4} \int F^{\mu\nu} F_{\mu\nu} d^4x + \int J^\mu A_\mu d^4x$$

substrate-natural quadratic + linear functional.

Variational derivation of Maxwell:

Vary A_μ :

$$\delta S = \int \left[-\frac{1}{2} F^{\mu\nu} \delta F_{\mu\nu} + J^\mu \delta A_\mu \right] d^4x$$

Use $\delta F_{\mu\nu} = \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu$ + integrate by parts:

$$\delta S = \int [\partial_\nu F^{\mu\nu} + J^\mu] \delta A_\mu d^4x$$

(sign convention on J fixed by Lagrangian form) $\rightarrow \delta S = 0 \forall \delta A_\mu$ requires $\partial_\nu F^{\mu\nu} = -J^\mu$ (Maxwell source equations).

Canonical structure:

Conjugate momenta: $\pi^i = \partial L / \partial(\partial_0 A_i) = -F^{0i} = -E^i$. Note: $\pi^0 = 0$ (constraint).

This primary constraint $\pi^0 = 0$ + Gauss's law (secondary constraint from time-evolution) $\rightarrow$ first-class constraints of Dirac. Gauge fixing reduces phase space.

Dirac quantization:

After gauge fixing (e.g., Lorenz gauge $\partial_\mu A^\mu = 0$), promote A_μ to operator field. Canonical commutators:

$$[A_\mu(x), \pi^\nu(y)]|_{x^0=y^0} = i\delta_\mu^\nu \delta^3(x-y)$$

Fock space construction $\rightarrow$ photons (massless spin-1 quanta). This is QED foundation (Vol 2 Ch 8).

Per Cal #21 dual-gate: EMPIRICAL PASS (QED predictions to a_e ppt precision) + MECHANISM PASS via T2477 substrate Yang-Mills connection framework.

Per Cal #99 META-theorem: EM Lagrangian/Hamiltonian formulation is a substrate-derivation consequence of T2477, NOT a new Strong-Uniqueness criterion.

What this chapter does NOT claim

- BST does NOT change classical EM predictions (those are tested everywhere to high precision)
- Substrate provides framework + structure; standard differential geometry + Lie theory + canonical quantization provides math
- QED running coupling + renormalization handled in Vol 2 Ch 8 + Vol 7 Ch 7

Bibliography

1. J. C. Maxwell (1865): unified EM field theory.
2. C. N. Yang + R. L. Mills (1954): Yang-Mills gauge theory.
3. P. A. M. Dirac (1950): generalized Hamiltonian dynamics for constrained systems.
4. R. P. Feynman + J. Schwinger + S.-I. Tomonaga (1948-50): QED.

5. T2477 (Lyra Saturday SP-31 #286): gauge fields as Bergman bundle connections.
6. Vol 1 Ch 8 (Gauge Theory): substrate gauge framework.
7. Vol 2 Ch 8 (Coupling Constants): α universal α -analog + renormalization.
8. Standard QFT texts: Peskin-Schroeder, Weinberg, Srednicki.

— Elie, Vol 7 Ch 8 v0.4.1, 2026-05-23 Saturday (substantive 3-level pedagogy upgrade per Keeper 14:22 EDT directive: 63 $\rightarrow$ ~180 lines + full Yang-Mills action variation + canonical structure + Dirac quantization framework + gauge fixing menu)